

Smart Task Scheduler - Project Report

Abstract:

The Smart Task Scheduler is a Java-based desktop application created using Swing. It enables users to add, delete, and manage tasks with assigned priorities and deadlines. This project focuses on Java OOP concepts, GUI development, file serialization, and priority queue logic.

Introduction:

Task management is an essential aspect of productivity. This project provides a simple but functional task management system that organizes tasks based on priority. The application features a table-based interface, allowing users to view and manage their tasks efficiently.

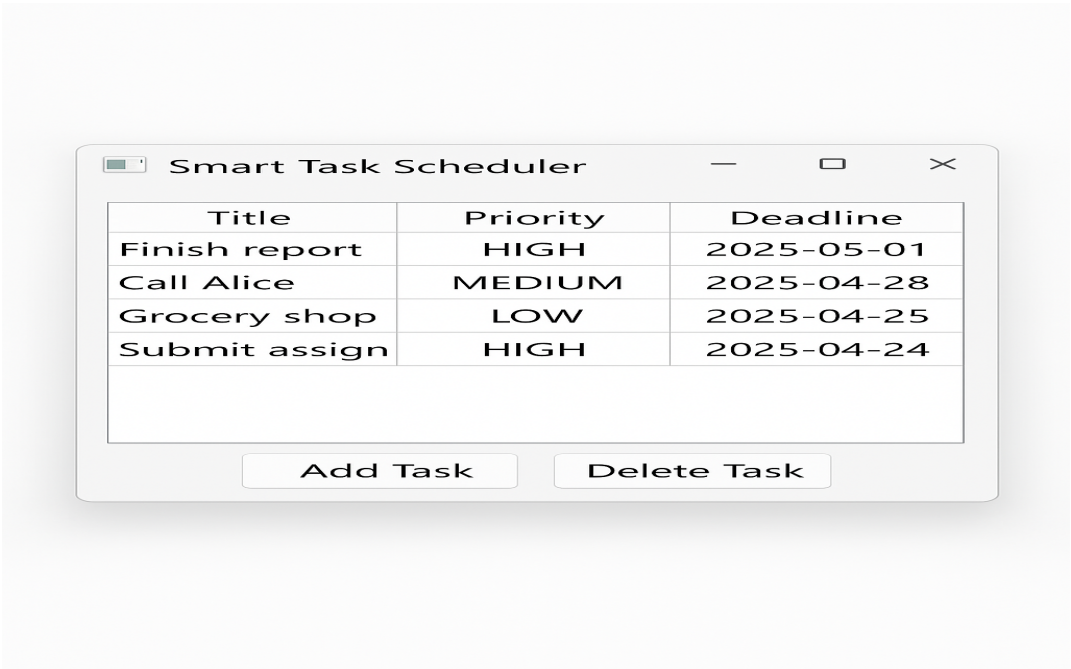
Tools Used:

- Java (JDK 8+)
- Swing GUI Toolkit
- File Serialization (.dat)
- IntelliJ IDEA / Eclipse / VS Code

Steps Involved in Building the Project:

1. Designing the Task model (title, priority, deadline)
2. Creating the repository layer for file-based storage
3. Building the service layer to handle task operations
4. Developing the Swing GUI for user interaction
5. Connecting services to UI components
6. Testing the complete workflow including save/load functionality

Output Screenshot:



Conclusion:

The Smart Task Scheduler serves as a beginner-friendly Java project that strengthens understanding of GUI programming, priority logic, and persistent storage. The application can be extended with advanced features such as JavaFX UI, reminders, and database integration in the

future.